

Genetic algorithm segmentation in partial least squares structural equation modeling

Christian M. Ringle · Marko Sarstedt ·
Rainer Schlittgen

Published online: 29 March 2013
© Springer-Verlag Berlin Heidelberg 2013

Abstract When applying the partial least squares structural equation modeling (PLS-SEM) method, the assumption that the data stem from a single homogeneous population is often unrealistic. For the full set of data, unobserved heterogeneity in the PLS path model estimates may result in misleading interpretations. This research presents the PLS genetic algorithm segmentation (PLS-GAS) method to account for unobserved

The authors presented the PLS-GAS method at the following conferences: 5th International Symposium of PLS and Related Methods, Ås, Norway, 2007; the 32th Annual Conference of the German Classification Society—Gesellschaft für Klassifikation (GfKl), Hamburg, Germany, 2008; the Australian & New Zealand Marketing Academy (ANZMAC) Annual Conference, Melbourne, Australia, 2009; and the Korean Academy of Marketing Science (KAMS) Global Marketing Conference, Tokyo, Japan, 2010.

Electronic supplementary material The online version of this article (doi:[10.1007/s00291-013-0320-0](https://doi.org/10.1007/s00291-013-0320-0)) contains supplementary material, which is available to authorized users.

C. M. Ringle
Institute for Human Resource Management and Organizations (HRMO),
Hamburg University of Technology (TUHH), Hamburg, Germany
e-mail: hrmo@tuhh.de
URL: <http://www.tuhh.de/hrmo>

C. M. Ringle · M. Sarstedt
Faculty of Law and Business, University of Newcastle, Newcastle, Australia

M. Sarstedt (✉)
Otto-von-Guericke-University Magdeburg, Magdeburg, Germany
e-mail: Marko.Sarstedt@ovgu.de
URL: <http://www.marketing.ovgu.de>

R. Schlittgen
Institute for Statistics and Econometrics, University of Hamburg, Hamburg, Germany
e-mail: rainer.schlittgen@uni-hamburg.de
URL: <http://www.wiso.uni-hamburg.de/index.php?id=schlittgen>

heterogeneity in the path model estimates. The results of a simulation study guide an assessment of this novel approach. PLS-GAS allows for uncovering unobserved heterogeneity and identifying different groups within a data set. In an application on customer satisfaction data and the American customer satisfaction index model, the method identifies distinctive group-specific PLS path model estimates which allow for a further differentiated interpretation of the results.

Keywords Genetic algorithm · Partial least squares · Path modeling · PLS-SEM · Segmentation · Structural equation modeling

1 Introduction

Structural equation modeling (SEM) is a well-established data analysis method in research and business practice that permits the estimation of complex cause–effect relationship models with latent variables. Covariance-based structural equation modeling (CB-SEM) (Jöreskog 1978, 1982) and partial least squares structural equation modeling (PLS-SEM) (Lohmöller 1989; Wold 1982) constitute the two key statistical techniques for estimating causal models (Dijkstra 1983; Fornell and Bookstein 1982; Hair et al. 2011; Jöreskog and Wold 1982). Whereas CB-SEM has long been the predominant approach for estimating structural equation models (Shah and Goldstein 2006), PLS-SEM has in recent years been increasingly disseminated in a variety of disciplines, including accounting (Lee et al. 2011), operations management (Peng and Lai 2012), management information systems (Ringle et al. 2012), marketing (Hair et al. 2012c), and strategic management (Hair et al. 2012a,b, 2013b; Robins 2012).

Applications of SEM in general and, more specifically, PLS-SEM usually analyze the full set of empirical data and, thus, implicitly assume that the used data stem from a single homogeneous population. This assumption of relatively homogeneous data characteristics is often unrealistic. Individuals (e.g., in their behavior) or corporations (e.g., in their structures) are different and pooling data across observations is therefore likely to produce misleading results (Jedidi et al. 1997; Wedel and Kamakura 2000). Consequently, the interpretation of PLS path model estimates may be ambiguous, incomplete, and ineffective if the heterogeneity of the respondents involves significantly different group-specific parameter estimates (Esposito Vinzi et al. 2008; Rigdon et al. 2010, 2011; Ringle et al. 2010).

Researchers who take heterogeneity into account frequently rely on a priori information to partition observations in respect of certain observable characteristics (e.g., socio-demographic information). However, while the consideration of observed heterogeneity generally proves valuable from a theoretical perspective, heterogeneity is often unobservable and cannot be attributed to any predetermined variable(s) (Jedidi et al. 1997).

To detect and treat unobserved heterogeneity, researchers frequently apply traditional clustering procedures (e.g., k -means) to manifest or latent variable scores. In structural equation modeling, this procedure proves problematic as it does not take the underlying structural model into account. Prior research recognized this weakness and proposed response-based segmentation techniques for detecting and treating

a discrete distribution of unobserved heterogeneity (Jedidi et al. 1997; Sarstedt and Ringle 2010). However, a review of these approaches reveals severe drawbacks which limit their applicability in many research situations (Sarstedt 2008a). Specifically, the key limitations of the existing approaches to segmentation in PLS-SEM lie in their inability to explore a sufficiently large fraction of the overall search space, to account for heterogeneity in Mode B measurement models of latent variables,¹ and to omit distributional assumptions to conform to the PLS-SEM methodology. An effective response-based segmentation approach in PLS-SEM should, however, avoid premature convergence to local optimum solutions and be capable of coping with model and data constellations commonly encountered in empirical research.

Overcoming the drawbacks of previous approaches, this paper builds on Ringle et al. (2013) to further develop a response-based segmentation method for PLS-SEM that effectively uncovers discrete distributions of unobserved heterogeneity. This method is also universally applicable across a wide range of model constellations. The proposed method combines a genetic algorithm (GA) with the PLS-SEM algorithm to estimate path models with latent variables. The partial least squares genetic algorithm segmentation (PLS-GAS) approach uses guided random searches to find an optimum solution in the complex search space that underlies segmentation tasks in PLS-SEM. Since PLS-GAS does not rely on distributional assumptions, it fits the PLS-SEM method's non-parametric character. Furthermore, it allows for the integration of both Mode A and Mode B measurement models, as defined by Wold (1982).

To evaluate the capabilities of the new approach, several computational experiments were carried out with artificially generated data groups. The data for the pre-specified PLS path models were generated with a newly developed data generation procedure that builds on the covariance matrix of the underlying path model pre-specifications. This type of approach delivers the data required for computational experiments with all the relevant factors for segmentation purposes (e.g., Mode A and Mode B measurement models with pre-specified loadings and weights). The results demonstrate the capabilities and effectiveness of PLS-GAS in various situations encountered in applications with empirical data. In addition, these results underline the approach's superior performance compared to two recently proposed response-based segmentation methods in PLS-SEM.

The remainder of this paper is organized as follows: in the next section, we review the existing approaches to segmentation in PLS-SEM. This review is followed by a description of how the PLS-GAS approach overcomes the shortcomings of existing methods. Thereafter, we use a newly developed data generation procedure for pre-specified PLS path models to conduct computational experiments that permit an assessment of the PLS-GAS performance across a wide range of data constellations. In addition, we illustrate the method's usefulness by applying PLS-GAS to the well-known American customer satisfaction index (ACSI) model. The paper concludes with a summary, a description of the study's limitations, as well as future research directions.

¹ Wold (1982) differentiates between Mode A and Mode B measurement models. In Mode A, the weight relations are from the latent variable to the associated manifest variables; in Mode B, the weight relations are from the manifest variables to the associated latent variable. In line with Wold (1982), we refer to Mode C path model constellations if each of Modes A and B was chosen at least once in the model.

2 Why do we need a new segmentation method?

Based on the seminal papers by [Jedidi et al. \(1997\)](#), [Hahn et al. \(2002\)](#) introduced the finite mixture partial least squares (FIMIX-PLS) approach—the first method for response-based segmentation in PLS-SEM. This approach uses the mixture regression concept to simultaneously assign observations to segments and estimate group-specific path coefficients. A particularly beneficial feature is the availability of segment retention criteria such as AIC, BIC and CAIC for model selection ([Sarstedt 2008b](#); [Sarstedt et al. 2011a](#)). While FIMIX-PLS has been used in several empirical studies ([Hahn et al. 2002](#); [Money et al. 2012](#); [Navarro et al. 2011](#); [Sarstedt et al. 2009](#)) and its properties have been researched in simulation studies ([Esposito Vinzi et al. 2007](#); [Ringle et al. 2013, 2010](#); [Sarstedt et al. 2011a](#)), it is not without problems. A primary concern relates to FIMIX-PLS requiring multivariate normally distributed data which runs contrary to the PLS-SEM technique's non-parametric nature. Especially in marketing and management where the data are rarely normal and more frequently highly skewed, this constraint can become problematic ([Squillacciotti 2010](#)) even though finite mixture methods are well known to be relatively robust against violations of its underlying distributional assumptions ([Esposito Vinzi et al. 2007](#)). Furthermore, FIMIX-PLS relies on the EM algorithm, which often converges to local maxima, especially when the number of estimated parameters is large, or the segments are not well separated ([Wedel and Kamakura 2000](#)). Lastly, as input for the mixture regressions, FIMIX-PLS builds on latent variable scores estimated for the full set of data, thereby keeping the measurement models constant across all segments to ensure convergence. Consequently, FIMIX-PLS is only capable of capturing heterogeneity in the structural model relationships and cannot account for heterogeneity in measurement models, which clearly limits its usefulness for empirical research settings.

To overcome some of FIMIX-PLS' limitations, [Squillacciotti \(2010, 2005\)](#) introduced the PLS typological path modeling (PLS-TPM) procedure, which [Esposito Vinzi et al. \(2007, 2008\)](#) have advanced by presenting the response-based procedure for detecting unit segments (REBUS-PLS). REBUS-PLS gradually re-allocates observations from one segment to the other with the goal of minimizing the residuals. Since the goal criterion is based on the residuals of the measurement models and the structural model (i.e., the GoF index; [Henseler and Sarstedt 2013](#)), REBUS-PLS takes the heterogeneity in the entire model into account. However, the goal criterion does not specifically allow accounting for heterogeneity in latent variables that use the Mode B measurement model setup ([Esposito Vinzi et al. 2008](#)). With Mode B measurement models' increasing importance in research ([Cenfetelli and Bassellier 2009](#); [Coltman et al. 2008](#); [Diamantopoulos 2011](#); [Diamantopoulos et al. 2008](#)), this limitation of REBUS-PLS is critical. Moreover, it is a gradient method that can only search very limited fractions of the overall search space. Identifying the best PLS-SEM segmentation result in respect of a certain goal criterion is a combinatorial data assignment problem whose complexity increases exponentially with higher numbers of observations and/or higher numbers of segments ([Cowgill et al. 1999](#)). By disregarding a vast number of potential solutions, REBUS-PLS offers only limited potential for uncovering solutions close to the global optimum. Furthermore, as in conventional clustering ([Maulik and Bandyopadhyay 2000](#); [Murthy and Chowdhury 1996](#)), data assignment

problems in PLS-SEM presumably have many local optimum solutions (Ringle and Schlittgen 2007). Gradient methods such as REBUS-PLS, however, suffer from premature convergence, since no progress can be made once a local optimum solution has been reached.

So far, FIMIX-PLS is the key segmentation method in PLS-SEM (Sarstedt 2008a). However, PLS-TPM and its improvement, REBUS-PLS, are two of the first non-parametric segmentation approaches specifically developed for PLS-SEM to uncover unobserved heterogeneity (see Sarstedt 2008a and Becker et al. 2013 for a comprehensive comparison of PLS-SEM segmentation methods). Although these methods have some limitations, we use them (i.e., FIMIX-PLS and REBUS-PLS) as benchmarks to evaluate the performance of the PLS-GAS method in computational experiments.²

3 Genetic algorithm segmentation in PLS-SEM

3.1 Problem characterization and method

In PLS-SEM, segmentation requires assigning data to the pre-specified number of K groups. This enables estimating the path model of each group of observations and determining the segmentation's quality with respect to a given objective optimization problem function. However, identifying the best PLS-SEM segmentation solution is an exceedingly complex combinatorial data assignment problem whose complexity increases exponentially with the number of observations and/or groups of data (Anderberg 1973; Cowgill et al. 1999; Murthy and Chowdhury 1996).

To handle this complexity, and drawing on Ringle et al. (2013), we propose a genetic search/hill-climbing hybrid (Cowgill et al. 1999) to detect and treat a discrete distribution of unobserved heterogeneity in PLS-SEM applications (Rigdon et al. 2010). As Wedel et al. (1999, p. 223) point out: "To a large extent, the issue of a continuous versus a discrete distribution of heterogeneity is an empirical one. A continuous heterogeneity distribution can be approximated closely by a discrete one by letting the number of support points of the discrete distribution increase at the cost of a decrease in the reliability of the parameters". Analyzing continuous heterogeneity distributions is known to have certain advantages, for example, in terms of better characterization of the tails of heterogeneity distributions, better prediction of individual behavior, and individual-level model estimates. However, discrete heterogeneous distributions, which are less sensitive to an assumed distribution of the parameters (e.g., normal), provide a finite mixture of homogeneous underlying segments. These distributions are therefore particularly useful to provide results that can be interpreted for managerial actions (e.g., for market segmentation) (Wedel et al. 1999). This especially holds for PLS-SEM which is routinely used to explain the variance of key target constructs in a model by different sources of explanation (e.g., success factors) (Albers 2010). Hence, revealing discrete segments and their segment-specific model estimates is particularly useful to develop recommendations on the actionable drivers for each group of data

² Note that Becker et al. (2013) recently presented another segmentation approach for PLS-SEM: PLS prediction-oriented segmentation (PLS-POS). The comparison of PLS-GAS and PLS-POS remains an issue for future research.

(Völckner et al. 2010). If an explanatory variable matches the discrete segmentation outcome very well, then researchers can further characterize the uncovered segments which improves the usefulness and specific relevance of results (Ringle et al. 2010; Sarstedt et al. 2009).

Initially, the approach explores the search space by using a non-deterministic GA (Michalewicz 1996) to provide a global near-optimal solution (first stage). However, since there is no guarantee that the GA will provide a global optimum solution, the best found segmentation result subsequently becomes the starting partition for a deterministic hill-climbing approach to locally improve (if possible) the GA outcome and to determine the final best partition (second stage) (Ringle et al. 2013).

3.2 Objective function

PLS-GAS aims at uncovering the segmentation solution which minimizes the error variance of the PLS-SEM segmentation results. Specifically, in both stages, PLS-GAS aims to minimize the objective function

$$F = \sum_{nml'k} \delta_{nml'k}^2 + \sum_{nml'k} \varepsilon_{nml'k}^2 + \sum_{nl^*k} \zeta_{nl^*k}^2, \quad (1)$$

with $\delta(\varepsilon)$ the residual term of the n th observation in the k th segment, corresponding to the m th manifest variable for the l' th exogenous (endogenous) latent variable with a Mode A measurement model; ζ is the residual structural model of the l^* th endogenous latent variable for the n th observation in the k th segment. These residual terms stem from PLS-SEM analyses of the groups as provided by PLS-GAS. Since the objective function F aims to maximize the model's explained variance per segment, it fits the prediction-oriented character of the PLS-SEM methodology (Hair et al. 2013a; Lohmöller 1989; Wold 1982).

3.3 First stage

The encoding of a PLS-SEM segmentation solution requires a vector (or chromosome) of N genes. Each position (or gene) n of the chromosome represents an allocation of the respective n th observation to one of the K segments. Consequently, an individual's chromosome represents a PLS-SEM segmentation solution to fit the overall data set's N observations into a pre-specified number of K segments. PLS-GAS starts by randomly generating a number of I individuals (i.e., PLS-SEM segmentation solutions). Based on this initial population, PLS-GAS probabilistically selects individuals to form new generations of populations. The probability of an individual being selected for the next generation is determined by the value of its objective function F (i.e., its fitness). Compared to all individuals in the population, the fitness F of an individual determines its rank r and its rank fitness $I + 1 - r$ for selection. Individuals with higher rank fitness have a better chance of building the next generation, which is determined by the selection probability $p_s = I(I + 1)/2$.

In GAs, selected individuals usually undergo one of the following three genetic operators: recombination (i.e., crossover), mutation, or reproduction (Holland 1975).

However, Fogel (2006) notes that the crossover operator negatively affects the algorithm's performance when used on non-uniform data structures. Since GA segmentation deals with non-uniform populations (e.g., data structures), we follow Fogel's (2006) recommendation and use a mutation-based search with an elitist selection (Murthy and Chowdhury 1996).

Thus, the genetic algorithm depends on mutation and recombination that apply with given probabilities p_m and p_r (with $p_m + p_r = 1$), respectively. A random number a ($0 \leq a \leq 1$) drawn from a uniform distribution, permits one of the two genetic operations to be chosen for each selected individual: mutation, if $a \leq p_m$, and reproduction if $a > p_m$. In the case of mutation, each gene in the selected individual's chromosome has a small transformation probability p_t . However, if the chromosome remains unchanged, the procedure ensures that at least one randomly selected gene mutates. In the case of reproduction, an unmodified copy of the selected individual becomes a member of the new generation. Independent of the choice of genetic operators, PLS-GAS retains the best found individual for the next generation (elitist selection).

A validity test ensures that each population has I distinctive individuals with valid segmentation solutions. More specifically, the validity test assesses whether each individual includes the minimum assignment of data points required for group-specific PLS path model computations. Invalid individuals are replaced until a population size of I individuals is reached. The first stage of PLS-GAS terminates after populating a fixed number of S generations. Figure 1 provides a brief description of the specific GA operational sequence in the first stage of the PLS-GAS algorithm.

In line with previous research (Fogel 2006), this study draws on the following parameter settings for PLS-GAS's first stage: (1) number of individuals in each population ($I = 100$); (2) probability of mutation: $p_m = 0.3$ and, accordingly, probability of reproduction: $p_r = 1 - p_m = 0.7$; (3) probability of transformation: $p_t = 0.1$; (4) number of generated populations (i.e., generations): 1,000.³

3.4 Second stage

There is no guarantee that the first stage discloses a local or even the global optimum solution. Hence, PLS-GAS selects the fittest individual from the first stage as the starting partition to locally improve the solution (if possible) in the second stage. Figure 2 provides a brief description of the second stage of the PLS-GAS methodology.

Following an idea proposed for PLS-TPM (Esposito Vinzi et al. 2007; Squillacciotti 2005) and REBUS-PLS (Esposito Vinzi et al. 2007, 2008) and its improved procedure suggested by Becker et al. (2009, 2013), local improvement investigates if an object's alternative cluster assignment further improves the value of the objective function. Initially, the first observation becomes a candidate for reassignment. Computation of the objective function F for this observation uses the PLS-SEM results of the segment to which the observation has been assigned. Subsequently, the procedure determines the F value of this observation for its potential assignment to each of the other segments.

³ To determine the number of individuals and generations, we examined these parameters' effects on the fitness. The results revealed only marginal effects of the population size for higher numbers of generations.

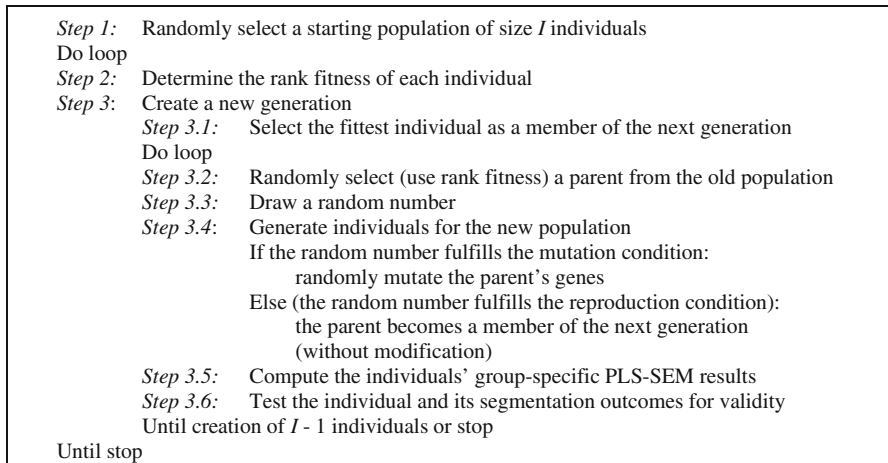


Fig. 1 First stage of PLS-GAS: genetic algorithm

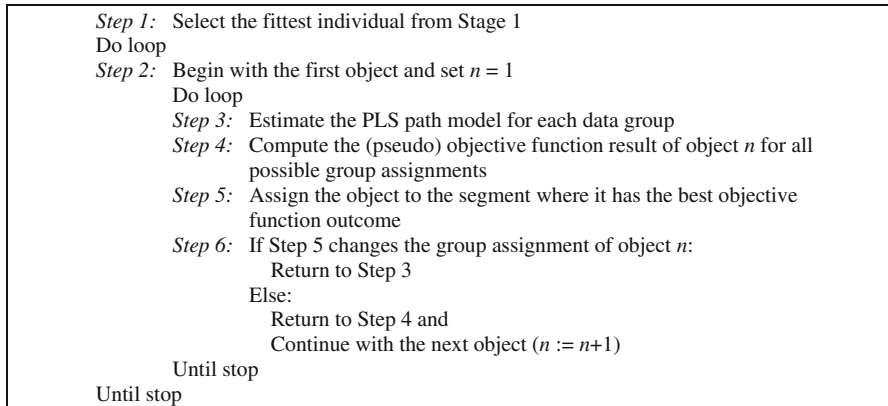


Fig. 2 Second stage of PLS-GAS: local improvement (hill-climbing)

This type of fast computation procedure employs the existing segment-specific path modeling results. Consequently, the procedure allows determining the pseudo-distance of each observation to its assigned group of data, and all possible alternative group assignments, without having to run the PLS-SEM algorithm in each case. The observation that is currently being evaluated becomes a member of that group of data to which it had the lowest pseudo distance. If the object remains in its original segment, the analysis continues with the next observation. Alternatively, after the observation's re-allocation, the approach updates the group-specific PLS path model estimates and continues to compute the pseudo-distances of the next observation. Local improvement stops when no relocation occurs in a single loop or—since convergence is not guaranteed—the algorithm reaches the maximum number of 30 iterations. PLS-GAS designates the solution with the minimum objective function value as the final best partition.

4 Analytical design of the computational experiments and data generation

The initial assessment of the PLS-GAS performance by means of computational experiments included factors well known to affect the estimates and the segmentation results in PLS-SEM. The identification of relevant factors and their different levels for this study was based on previous simulation studies on CB-SEM (Boomsma and Hoogland 2001; Reinartz et al. 2009), PLS-SEM (Chin 2003; Henseler and Chin 2010), and segmentation studies on PLS-SEM (Becker et al. 2013; Esposito Vinzi et al. 2007; Hahn et al. 2002; Sarstedt et al. 2011a).

Data

- Number of observations [50, 100, 150].
- Data distribution [normal, non-normal].⁴

Measurement model

- Number of manifest variables in the equally sized measurement models of the three latent variables (Anderberg 1973; Aptech 2008; Boomsma and Hoogland 2001).
- Loadings and weights of the manifest variables [high, low, mixed].
We differentiate three error variance situations—low loadings (i.e., all outer loadings have a standardized value of 0.75), loadings from low to high (i.e., the outer loadings per measurement have a standardized value that linearly increases from 0.75 to 0.95), and high loadings (i.e., all outer loadings have a standardized value of 0.95)—for Mode A measurement models.⁵
- PLS path model constellations [Mode A; Mode C].
Mode A refers to path models which only comprise Mode A measurement models; Mode C refers to path models which contain both, Mode A and Mode B measurement models.

Structural model

- The error variance of the endogenous latent variable [5, 10, 20 %] and, thus, the separability of data in terms of the difference between the group-specific structural model relationships D .⁶
- PLS path model complexity as depicted in Fig. 3 [simple path model, complex path model].

⁴ In this study, we generate highly non-normal data with a skewness of 6.183 and an excess (resulting from the difference between two log-normal distributions) of 33.067. See the Online Appendix II (<http://www.pls-sem.com/orsp/oa.pdf>) for further information on the data generation procedure.

⁵ Although such changes in Mode B measurement models do not change the error variances, the pre-specified weights of these measurement models are changed similarly—as in Mode A measurement models—in the computational experiments. When the weights are low (high), they have a pre-specified value of 0.20 (0.40). The mixed factor level assigns the high pre-specified weight of 0.40 to the first manifest variable and the low weight of 0.20 to the last manifest variable per Mode B measurement model; in the mixed factor level constellation, weights linearly increase from 0.20 to 0.40.

⁶ See Equation 16 in Online Appendix II (<http://www.pls-sem.com/orsp/oa.pdf>).

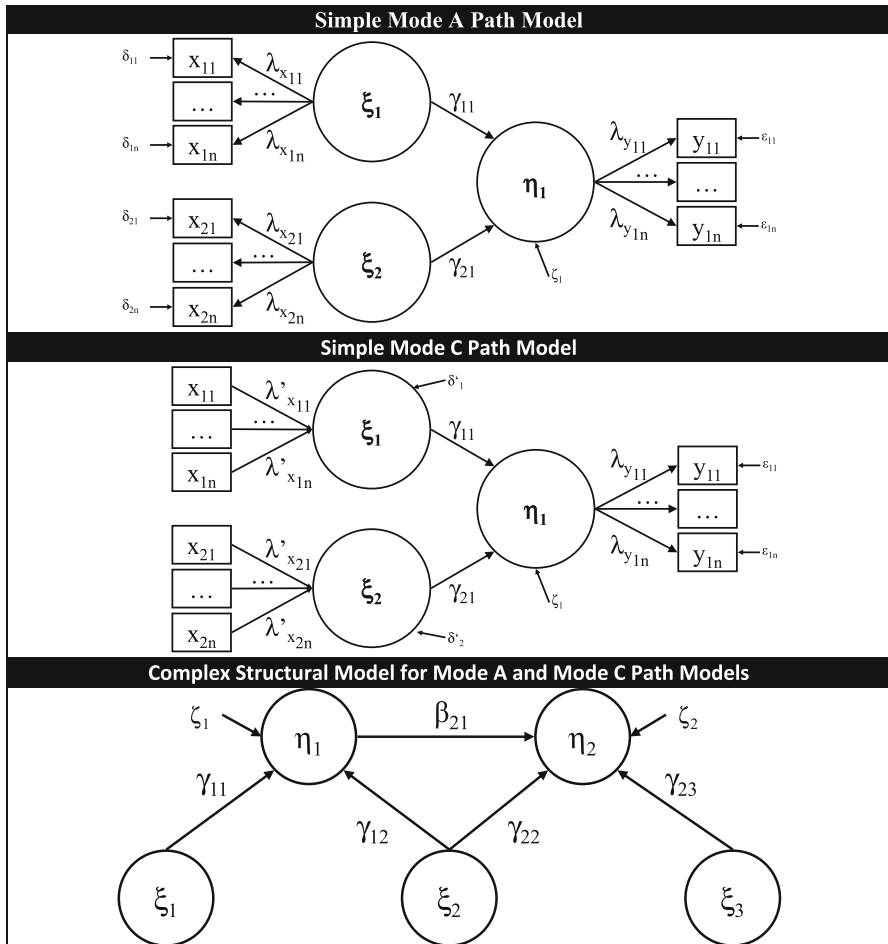


Fig. 3 PLS path models (Table 1 in Online Appendix I (<http://www.pls-sem.com/orsp/oa.pdf>) provides an explanation of all used symbols)

Segment characteristics

- The number of segments (Albers and Hildebrandt 2006; Anderberg 1973).
- The relative segment sizes of the segments [balanced, unbalanced].⁷

Changes in the number of observations and the number of indicators per measurement model refer to estimated PLS path model coefficients' well-known consistency at large property (Jöreskog and Wold 1982). Alternative levels of loadings and weights in the measurement models, the endogenous latent variables' error variance, the difference between group-specific structural model coefficients, the altered

⁷ For two segments, relative segment sizes are 50 %/50 % (balanced) and 75 %/25 % (unbalanced); for three segments, relative segment sizes are 33 %/33 %/33 % (balanced) and 60 %/20 %/20 % (unbalanced).

relative segment sizes, additional groups, complex models, and non-normal data situations aim at increasing the difficulty of separating the pre-specified sets of data (Sarstedt et al. 2011a). Hence, these factors are particularly relevant for analyzing PLS-GAS's performance. Finally, we included Mode C path model constellations to analyze the general PLS-GAS applicability to alternative types of PLS path models (i.e., modes of measurement models). The resulting design is factorial. We conduct 30 replications (data sets) for each cell to obtain stable average outcomes for our analysis. Hence, we compute PLS-GAS results for a total number of $2^5 \times 3^4 \times 30 = 77,760$ data sets.

Data generation for the pre-specified factor-level combinations uses the procedure introduced in Online Appendix II (<http://www.pls-sem.com/orsp/oa.pdf>). This method is the first to transfer the PLS-SEM specifications into a covariance matrix and to generate data accordingly (closed-form PLS-SEM model representation and data generation). This novel method for generating data of pre-specified PLS path models has been extended to also generate data for pre-specified formative measurement models. Hence, Mode B and Mode C setups PLS-SEM can be analyzed. As shown in Online Appendix II (<http://www.pls-sem.com/orsp/oa.pdf>), the procedure generates data that precisely meet the model pre-specifications (e.g., structural model path coefficients, formative weights, reflective loadings, or R^2 values).⁸

The generation of the artificial sets of data for the computational experiments and the PLS-GAS runs were carried out in GAUSS 9.0 (Aptech 2008).⁹ In addition to a stand-alone assessment of PLS-GAS, we also compared the approach's performance with that of FIMIX-PLS and REBUS-PLS. FIMIX-PLS was run with the flexmix package (Grün and Leisch 2008) using the latent variable scores obtained from prior PLS path model estimations with semPLS (Leisch and Monecke 2012) as input; REBUS-PLS was run using the plspm package (Sánchez and Trinchera 2013). The flexmix, semPLS and plspm packages have been developed for the statistical software application R (R Core Team 2012). For the parameter specifications, we used the default settings recommended by developers of the flexmix, semPLS and plspm packages.

5 Results

5.1 Core set of analyses

Two factors, the number of observations and the number of indicators per (Mode A) measurement model, are well known to have an impact on PLS path model estimates. Simulation studies (Chin et al. 2003; Reinartz et al. 2009) reveal a PLS-SEM bias of results (i.e., over-estimation of the outer relationships and underestimation of the inner relationships) when the number of observations and/or the number of manifest variables in Mode A measurement models is relatively small. Increasing these two factors at the same time reduces the PLS-SEM bias of path model estimates (i.e., the

⁸ The data sets that have been used as input for the analyses are available at <http://www.pls-sem.com/orsp/data.zip>.

⁹ The GAUSS program to run PLS-GAS is available at <http://www.pls-sem.com/orsp/program.zip>.

consistency at large property) (Hui and Wold 1982). Our computational experiments show that neither

- the altered number of observations, nor
- the number of manifest variables per measurement model

have a significant effect on PLS-GAS's performance. Hence, in light of the study's complexity, we refrain from presenting all outcomes but focus on those factors that have an effect on PLS-GAS's performance. We continue presenting results for 50 observations per group of data and five indicators per measurement model when analyzing additional factors that may affect PLS-GAS performance.

The results of the study substantiate negative effects of two key factors on PLS-GAS's performance:

- an increased level of the error variance for manifest variables in Mode A measurement models and/or the endogenous latent variables [i.e., $\text{Var}(\zeta)$],¹⁰ and
- lower differences in group-specific PLS path coefficients, as defined by the distance measure D ,¹¹ which entails smaller differences between the group-specific PLS-SEM estimates for structural path relationships in the model.

In each of the following presentations of results, these factors are combined with an additional relevant factor of analytic interest for segmentation purposes (e.g., the number of groups) to provide a clear picture of PLS-GAS's performance which is contrasted with those of FIMIX-PLS and REBUS-PLS.

Each combination includes nine graphics in a 3×3 matrix arrangement. The rows represent the different specifications for the outer loadings (and weights), while the columns include alternative settings for the error variance of endogenous latent variables in the structural model. In each graphic, the x -axis entails increasing levels of the relative distance D of path coefficients between the groups; the y -axis presents the 95 % confidence interval of hit rates of PLS-GAS, FIMIX-PLS, and REBUS-PLS (i.e., the proportion of observations generated for a specific group and finally assigned to that group).

Figure 4 shows the results of the simple Mode A path model constellation, five indicators per measurement model, two groups of normally distributed data with 50 observations each. While the upper curves of the PLS-GAS and FIMIX-PLS results change with the different data situations of the factor-level combinations, REBUS-PLS in no way captures these changes in the data constellations. In Fig. 4, the lower curve of the REBUS-PLS results remains at a stable level with relatively low performance. More specifically, REBUS-PLS's performance always remains at a hit rate of about 50 % which is equivalent to a random assignment of data for two groups. These results for REBUS-PLS parallel those of Ringle et al. (2013) whose study also disclosed considerable weaknesses in the method's performance.

With regard to PLS-GAS's and FIMIX-PLS's performances, changed levels of loadings and, thus, different levels of indicator reliability do not affect their performances (see the development of the upper curves from the upper row to the lower row in Fig. 4).

¹⁰ Note that PLS-GAS always meets the global optimum solution when no error is present.

¹¹ See Equation 16 in Online Appendix II (<http://www.pls-sem.com/orsp/oa.pdf>).

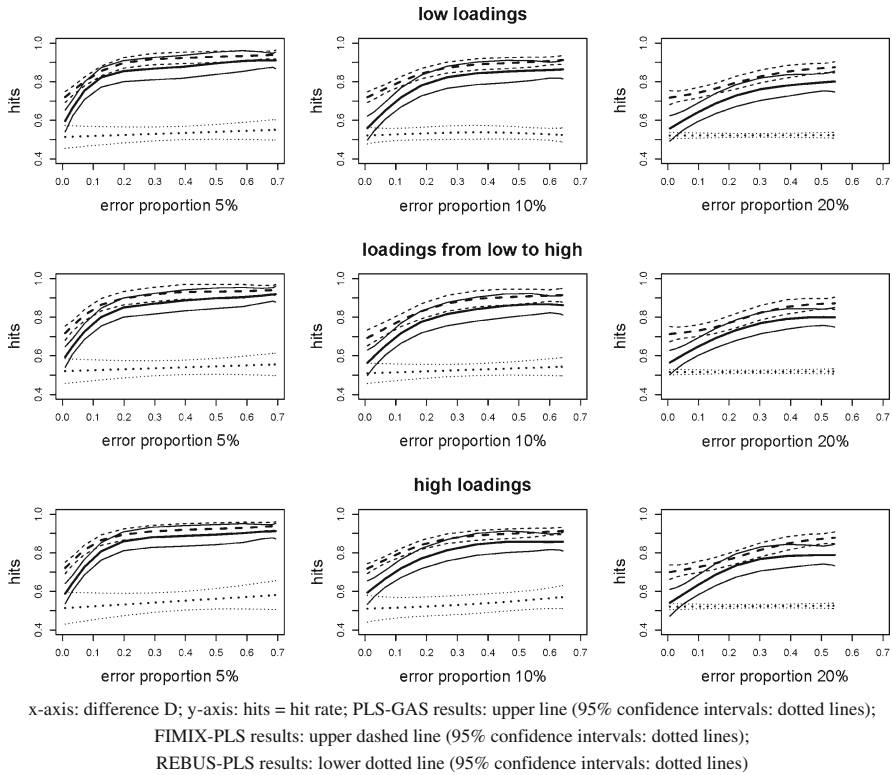


Fig. 4 Segmentation results of PLS-GAS, FIMIX-PLS, and REBUS-PLS (simple model, two groups with 50 observations per group; normal data)

On the contrary, changes in the structural model's error variance levels do have a significant impact. Figure 4 demonstrates that PLS-GAS's and FIMIX-PLS's performances decrease with higher levels of endogenous latent variables' error variance (see the development of the curves from the left column to the right column). Finally, higher differences D between group-specific PLS path coefficients estimates in the structural model entail strongly improved performances of PLS-GAS and FIMIX-PLS (x -axis in Fig. 4).

Regarding the hit rates, FIMIX-PLS seems to slightly outperform PLS-GAS. To gain deeper insight into the performance of the methods, we consider the simple model (Fig. 2) and a two-segment solution and compare the a priori determined group-specific coefficients that result from the artificially generated sets of data (i.e., γ_{11} , γ_{21} , γ_{12} , and γ_{21}) with the estimated values (i.e., $\hat{\gamma}_{11}$, $\hat{\gamma}_{21}$, $\hat{\gamma}_{12}$, $\hat{\gamma}_{21}$) for each group. The difference between the estimated group-specific coefficients (γ) and the a priori determined values (γ) represents the bias of the segmentation results. This bias is displayed on the y -axis in Fig. 5, while the x -axis entails increasing levels of the relative distance D of path coefficients between the groups. Every column in Fig. 5 includes the results for a specific coefficient (i.e., $\hat{\gamma}_{11}$, $\hat{\gamma}_{21}$, $\hat{\gamma}_{12}$, or $\hat{\gamma}_{21}$); the different rows represent the results for alternative error variance proportions of the endogenous latent variables (i.e., 5, 10,

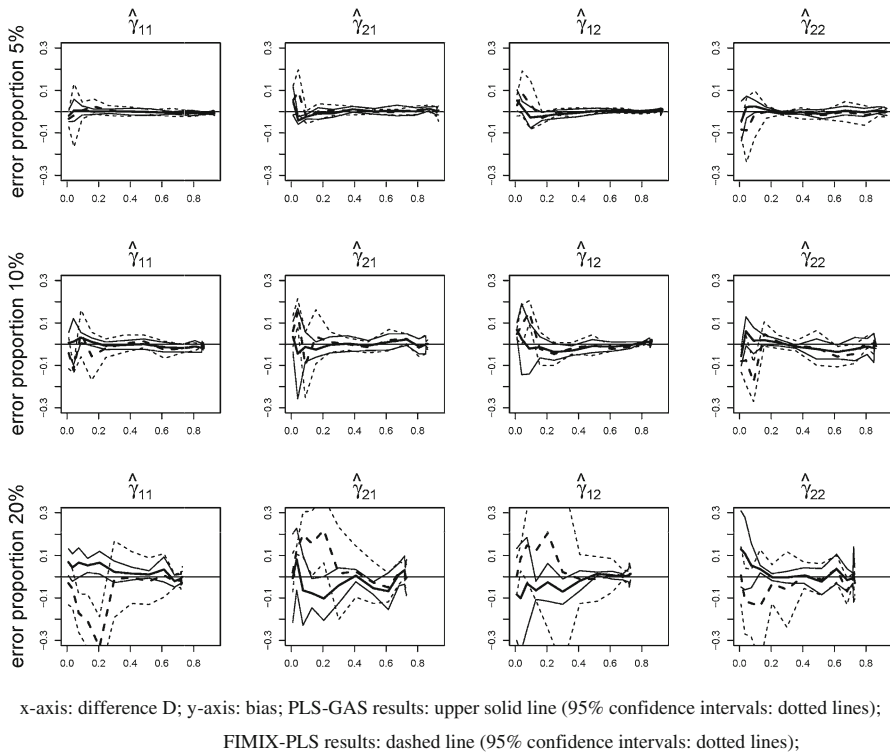


Fig. 5 Bias of PLS-GAS's and FIMIX-PLS's segmentation results (simple model, two groups with 50 observations per group; normal data)

and 20 %). Note that in the following, REBUS-PLS results are not considered as they entail large biases in the group-specific parameter estimates across all constellations, which substantiates the method's poor performance in terms of hit rates.

Even though, PLS-GAS's hit rates performance is slightly lower than that of FIMIX-PLS, the results in Fig. 5 show that PLS-GAS generally performs superior regarding the precision of the resulting segment-specific model estimates. In situations with intermediate and high levels of error variance (regardless of the distance D), PLS-GAS performs considerably better in terms of parameter bias than FIMIX-PLS. Only when error variances are low, there is practically no difference between the two methods in terms of parameter recovery.

Compared to FIMIX-PLS, the contradictory result for PLS-GAS (i.e., lower hit rates but higher precision for intermediate and high error variances) is not surprising. The artificially generated data sets have pre-specified levels of error. In accordance, observations exist that do not necessarily precisely fit into the one group or the other group of data as expected from the generation of artificial data. The classification of these data to one or the other segment results in minor changes in the objective criterion for grouping data, as well as in insignificant changes in the group-specific PLS-SEM estimates. PLS-GAS seems to assign these indistinct observations that result from

the pre-specified error variance levels to segments in accordance with its objective function and, thus, obtains higher precision, especially in situations with high levels of error variance. Comparing, for example, the lower right results representation in Fig. 4 with the lower right results representation in Fig. 5 substantiates this argument.

For larger data sets (e.g., 100 observations per group of data), the PLS-GAS hit rates increase and outperform those of FIMIX-PLS (Fig. 1 in Online Appendix II). While the FIMIX-PLS bias remains at comparable levels, for larger sets of data, the precision of PLS-GAS slightly increases in line with the improved hit rates (Fig. 2 in Online Appendix II) and, thus, PLS-GAS's superior performance becomes more pronounced, especially for data constellations with high levels of error variance.

5.2 Unbalanced segment sizes

Another factor, which is particularly relevant for practical applications, is the relative segment size. A segmentation method for PLS-SEM must group data effectively when the segments are not balanced in terms size. Figure 3 in the Online Appendix II presents the computational experiment results of the previous factor constellation (i.e., simple model, two groups, and normal data), which now have very unbalanced relative segment sizes of 25 % (50 observations) and 75 % (150 observations).

We find that the PLS-GAS and FIMIX-PLS performances are not affected by this factor. The outcomes regarding two relatively unbalanced segments of data (Fig. 3 in Online Appendix II) remain at the same level as for situations with balanced relative segment sizes (Fig. 4). The same kind of finding holds for the bias (Fig. 4 in the Online Appendix II) in the unbalanced relative segment size constellations, which is also very similar to the results displayed for the balanced segment situation displayed in Fig. 5.

We once again selected a factor-level constellation with relatively low numbers of observations for Figs. 3 and 4 in Online Appendix II. With larger numbers of observations, FIMIX-PLS again remains at comparable levels while the performance of PLS-GAS slightly improves.

5.3 Non-normal data

The non-parametric PLS-SEM approach does not require normally distributed data (Lohmöller 1989; Wold 1982). This aspect is important since most data collected in behavioral research do not follow a normal distribution (Micceri 1989). Hence, PLS-SEM is often used in situations when data are not normally distributed (Hair et al. 2011). Our study therefore analyzes how non-normal data affect PLS-GAS's performance. Figure 5 in the Online Appendix II shows the results for non-normal data when using the same factor constellations as for normal data. Compared to normal data, we find that PLS-GAS now also outperforms FIMIX-PLS regarding the hit rates across all data constellations.

In the same line, the advantages of PLS-GAS compared to FIMIX-PLS also increase for the precision of results in non-normal data constellations (Fig. 6 in Online Appendix II). PLS-GAS performs particularly well in situations where the error variances are rather low and the difference between the segment-specific coefficients

per group of data is relatively high. However, with increasing error variance and path differences, the performance of PLS-GAS decreases at a much faster rate than with normal data. Although PLS-GAS still performs very well with non-normal data, this factor has a clear negative impact on the method's performance. FIMIX-PLS's performance is equally affected by an increase in the difference between segment-specific coefficients and error variances. For intermediate levels of error variances, PLS-GAS clearly outperforms FIMIX-PLS, making it the preferred method for the analysis of non-normal data.

5.4 Additional groups of data and higher model complexity

Another concern is the effect of additional groups of data on PLS-GAS's performance. With additional groups, the data assignment problem becomes much more complicated. In accordance, one expects the performance of the segmentation methods for PLS-SEM to decrease in situations with additional numbers of pre-specified groups. Fig. 7 in Online Appendix II shows the PLS-GAS, FIMIX-PLS, and REBUS-PLS results for three segments and the same factor constellations used to obtain the results presented in Fig. 4. Our observation that the REBUS-PLS benchmark performs slightly above random data assignment is further substantiated by the results for this factor constellations. With three equally sized groups of data, the hit rate of REBUS-PLS declines to a level slightly above 33 % (Fig. 7 in Online Appendix II). For PLS-GAS and FIMIX-PLS, one also generally finds lower hit rates for the three groups of data compared to the two-group case. For example, a hit rate of about 70 % can only be reached with relatively low error variances of the endogenous latent variables and a relatively high distinctiveness of the group-specific PLS path coefficients. Thus, a higher number of groups decreases PLS-GAS's and FIMIX-PLS's hit rate performance. In fact, in this study's computational experiments, PLS-GAS and FIMIX-PLS can only provide satisfactory hit rates when the data are well separated (i.e., high difference D of group-specific path coefficients; low error proportion) across the three groups. Comparing the performance of PLS-GAS and FIMIX-PLS yields practically no differences in terms of parameter recovery.

In this context, we also analyzed the performance of PLS-GAS with regard to an ill-specified number of segments (i.e., over- and under-segmentation). Such situations negatively affect the correct assignment of observations to the pre-specified groups of data. However, the results clearly show that—despite the ill-specification—PLS-GAS does not randomly assign observations to the segments. The extracted groups of data have very high R^2 values and substantial relative segment sizes.¹²

Finally, the analysis considered a complex path model (Fig. 3) adopted from the studies by Albers and Hildebrandt (2006) and Gudergan et al. (2008). In addition to the relationships between exogenous and endogenous latent variables, this PLS path model also includes a relationship between endogenous latent variables. Compared to the simple path model (Fig. 4), we find that PLS-GAS performs consistently better with the

¹² See the Online Appendix IV (<http://www.pls-sem.com/orsp/oa.pdf>) for the results of these additional analyses.

more complex models considered in this study, regardless of whether they only include Mode A measurement models or also Mode B measurement models (Fig. 8 in Online Appendix II). This result contradicts our expectations (i.e., we envisaged lower PLS-GAS performances in situations that involve more complex PLS path models). Perhaps, a greater number of parameters offers more information for PLS-GAS to adequately partition the data into the segments, overcompensating the increased complexity of the segmentation task. However, an in-depth analysis regarding this outcome of our study remains a task for future research.

5.5 Mode B measurement models

Besides Mode A measurement models, PLS path models often include latent variables with Mode B measurement models (Hair et al. 2012b,c; Ringle et al. 2012). Thus, PLS-GAS's capability to be applicable in such situations (i.e., Mode C path model constellations) is an important issue for the method's general usability in empirical research. Figure 9 in Online Appendix II shows the results of our computational experiments as described for the data constellations in Fig. 4 but using Mode B measurement models for the exogenous latent variables (i.e., Mode C path model constellations). Since the REBUS-PLS and FIMIX-PLS methods cannot cope with Mode B measurement models (Chin et al. 2003; Peng and Lai 2012), we focus on presenting the hit rates of the PLS-GAS approach. Compared to model constellations that only include Mode A measurement models (Fig. 4), PLS-GAS performs worse in the Mode C path model constellation. Only in situations with low error variances and high differences between the group-specific path coefficients, PLS-GAS offers relatively high hit rates.

In the same line, the precision of group-specific PLS-GAS estimates (Fig. 10 in Online Appendix II) declines compared to the Mode A situation. In spite of Mode B measurement models having a negative effect on PLS-GAS's overall performance, the method shows capabilities for dealing with this kind of data constellation. PLS-GAS is capable of revealing underlying groups of data in Mode C PLS-SEM settings but at lower performance levels and, thus, with lower precision.

In contrast, the results for REBUS-PLS and FIMIX-PLS confirm that both methods have not been designed for formative measurement model constellations. REBUS-PLS is at similar low levels as described before and FIMIX-PLS also creates segments that are extremely different from the pre-specified sets of data (i.e., relatively large bias and extremely huge variance of results). Hence, PLS-GAS represents an important step towards providing segmentation methods for PLS-SEM that can also deal with formative measurement models.

6 ACSI model application

The ACSI model (Anderson and Fornell 2000; Fornell et al. 1996) is one of the most salient satisfaction models in business research and a very prominent PLS-SEM application. Furthermore, prior research (Rigdon et al. 2011; Ringle et al. 2010) has used FIMIX-PLS on the ACSI model and original data, making it particularly suitable

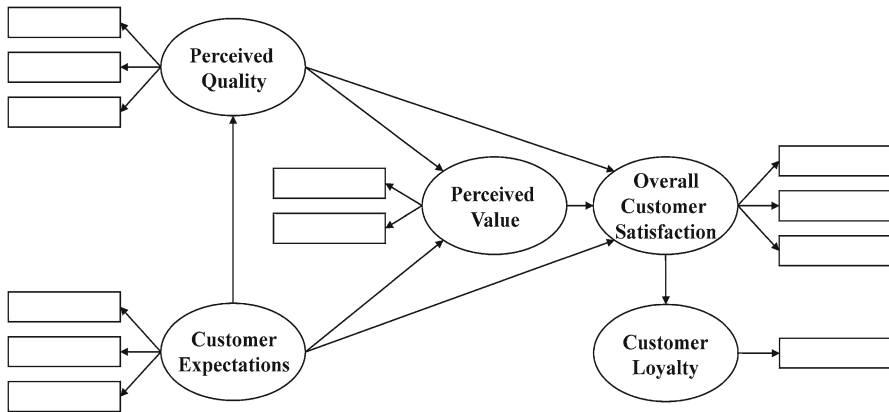


Fig. 6 ACSI model application

for the present study setting. [Rigdon et al. \(2011\)](#) and [Ringle et al. \(2010\)](#) provide for further explications of the ACSI model (Fig. 6) and the data used in this study.¹³

Parameter estimation is carried out using SmartPLS 2.0 ([Ringle et al. 2005](#)). An initial evaluation of the PLS-SEM estimates follows the suggestions provided by [Hair et al. \(2011, 2012c, 2013a,b\)](#),¹⁴ which reveal that the aggregate PLS-SEM analysis provides reliable and valid measures.

The remaining issue is whether distinct groups of customers exhibit significantly different parameter estimates. PLS-GAS allows this critical question to be answered. When applying the method, the researcher has to specifically specify the number of segments to be retained from the data set. However, PLS-GAS does not provide a clear indication of how to choose an appropriate number of segments. Instead of merely reverting to proxies, such as the solution's interpretability or acceptable R^2 values, we used FIMIX-PLS's segment retention criteria results for this empirical example to guide the decision ([Ringle et al. 2010](#); [Sarstedt et al. 2011a](#)). To find further support for the choice of the number of segments, we ran PLS-GAS for a two- to six-segment solution and compared the weighted average R^2 values and the differences of the segment-specific path coefficients. These analysis results clearly indicated that the two-segment solution is most appropriate in terms of uncovering latent segments that are differentiable. Table 1 presents the global model, as well as the PLS-GAS results in respect of two segments; the results of the FIMIX-PLS benchmark come from [Ringle et al. \(2010\)](#).

¹³ The data were provided by Fornell, Claes. American Customer Satisfaction Index, 1999 [Computer file]. ICPSR04436-v1. Ann Arbor, MI: University of Michigan. Ross School of Business, National Quality Research Center/Reston, VA: Wirthlin Worldwide [producers], 1999. Ann Arbor, MI: Inter-University Consortium for Political and Social Research [distributor], 2006-06-09. We would like to thank Claes Fornell and the ICPSR for making the data available.

¹⁴ In addition to these recommendations, one finds various recommendations on the use of the PLS-SEM method and the evaluation of results in prior literature; examples include: [Chin \(1998, 2010\)](#), [Falk and Miller \(1992\)](#), [Götz et al. \(2010\)](#), [Haenlein and Kaplan \(2004\)](#), [Henseler et al. \(2009, 2012\)](#), [Lohmöller \(1989\)](#), [Roldán and Sánchez-Franco \(2012\)](#), [Sosik et al. \(2009\)](#), [Tenenhaus et al. \(2005\)](#).

Table 1 ACSI model results

	Full data set	PLS-GAS			FIMIX-PLS		
		$k = 1$	$k = 2$	ldiff ^a	$k = 1$	$k = 2$	ldiff ^a
Customer Expectations of Quality → Perceived Quality	0.556***	0.788***	0.476***	0.312***	0.807***	0.258***	0.549***
Customer Expectations of Quality → Perceived Value	0.072***	0.708***	-0.225***	0.933***	0.218***	-0.107***	0.325***
Customer Expectations of Quality → Overall Customer Satisfaction	0.021**	0.094***	-0.025**	0.118***	0.117***	-0.068***	0.185***
Perceived Quality → Overall Customer Satisfaction	0.557***	0.426***	0.669***	0.243***	0.425***	0.633***	0.208***
Perceived Quality → Perceived Value	0.619***	0.090***	0.792***	0.702***	0.582***	0.544***	0.038**
Perceived Value → Overall Customer Satisfaction	0.394***	0.464***	0.286***	0.178***	0.455***	0.308***	0.147***
Overall Customer Satisfaction → Customer Loyalty	0.687***	0.745***	0.720***	0.025**	0.839***	0.481***	0.358***
	Full data set	$k = 1$	$k = 2$	Weighted average R^2	$k = 1$	$k = 2$	Weighted average R^2
Relative segment size	1.000	0.488	0.512		0.673	0.327	
R^2 Perceived Quality	0.309	0.620	0.227	0.419	0.651	0.067	0.460
R^2 Perceived Value	0.439	0.609	0.508	0.557	0.591	0.277	0.488
R^2 Overall Customer Satisfaction	0.777	0.792	0.775	0.783	0.848	0.679	0.793
R^2 Customer Loyalty	0.471	0.554	0.518	0.536	0.704	0.231	0.549

*** Sig. at 0.01, ** Sig. at 0.05, * Sig. at 0.1

^a PLS-SEM multi-group comparison, the nonparametric confidence set approach has been applied (Sarstedt et al. 2011b)

PLS-GAS reveals two segments of almost equal sizes. Segment-specific reliability and validity analyses reveal that all construct measures exhibit a high degree of internal consistency and meet all relevant measurement model evaluation criteria (Hair et al. 2011). Further analysis shows that all of the first segment's endogenous constructs achieve (considerably) higher R^2 values than the global model, thus indicating an increased predictive power (Table 1). Moreover, most R^2 values in the second segment are at similar levels when compared to the aggregate data analysis results. Calculating the overall R^2 as the sum of each endogenous construct's R^2 values across the two segments weighted by the relative segment size, shows that, on the whole, data segmentation could significantly increase the predictive power compared to the results of the overall data set. For example, while the *Loyalty* construct has an R^2 value of

0.471 for the overall data set, both group-specific R^2 values are higher (i.e., 0.554 and 0.518), resulting in a weighted average R^2 value of 0.536.

When comparing the global model with the results derived from PLS-GAS, one finds that the relative importance of the driver constructs differs quite substantially in the two segments (Table 1). While segment 1 has particularly strong effects from *Customer Expectations of Quality* to *Perceived Quality* and to *Perceived Value*, these relationships are considerably lower and even negative in the second segment. Hence, the importance of the customer expectations construct is far more pronounced in the first segment. Here, customers' expectations prior to purchase strongly influence all subsequent constructs, including the *Overall Customer Satisfaction* through the indirect relationships via *Perceived Quality* and *Perceived Value*. In contrast, the influence of *Perceived Quality* and *Perceived Value* on *Overall Customer Satisfaction* is stronger in the second segment than in the aggregate data analysis and in segment one. Overall, these results clearly suggest that the data are heterogeneous, which the PLS-GAS procedure can successfully reveal.

The nonparametric confidence set approach to PLS-SEM multi-group comparison (Sarstedt et al. 2011b) further substantiates these findings. The test results provide evidence that all paths related to the customer expectations construct differ significantly between $k = 1$ and $k = 2$ (Table 1). This reflects the varying importance of expected quality in respect of all the following constructs (most notably overall customer satisfaction), which provides the necessary grounds for the development of differentiated marketing and management activities.

In comparison, FIMIX-PLS uncovers one larger segment with a relative size of 0.673 and a smaller one with a relative size of 0.327. The method offers weighted average R^2 values similar to those of PLS-GAS. However, the larger segment has consistently higher R^2 values compared to the overall data set, while the smaller segment has consistently lower R^2 values. In contrast, the PLS-GAS results only have one R^2 value (i.e., *Perceived Quality* in segment 2) that is smaller compared to the results of the overall data set. Additional discussions and conclusions regarding the FIMIX-PLS results are provided by Ringle et al. (2010).

The results of this ACSI model application show that both PLS-GAS and FIMIX-PLS uncover groups of data with significantly different group-specific PLS path model estimates and higher weighted average R^2 values compared to the full data set. While FIMIX-PLS identifies one segment that offers superior results compared to the overall data set (in terms of R^2 values), PLS-GAS reveals two segments which are favorable in this respect. The latter finding is particularly important to conclude that the PLS-GAS results are advantageous in this empirical application.

7 Conclusions and future research

Given that heterogeneous data structures are more the rule than the exception (Mooi and Sarstedt 2011), uncovering unobserved heterogeneity is a critical issue that should be addressed in any PLS-SEM analysis (Hair et al. 2012c). Researchers and practitioners need effective means of segmentation in PLS-SEM. Drawing on Ringle et al. (2013), this article contributes to existing literature by presenting PLS-GAS as a

new PLS-SEM segmentation method designed to offer key improvements, especially with respect to existing methods' weakness (i.e., FIMIX-PLS and REBUS-PLS). The study presents a novel procedure to generate data for pre-specified PLS path models, including both Mode A and Mode B measurement models to test and compare the performance of PLS-GAS against the FIMIX-PLS and REBUS-PLS benchmarks. Computational experiments with a variety of key factors for segmentation purposes provide a comprehensive evaluation of the PLS-GAS methodology and reveal its capabilities and effectiveness when treating unobserved heterogeneity in path models. PLS-GAS performs consistently well in all combinations of

- numbers of observations,
- numbers of manifest variables in the measurement models,
- relative segment sizes,
- levels of outer loadings, and
- regardless of whether the data are normal or non-normal.

However, an analysis of our results reveals that

- a decreased distinctiveness in the segments,
- the error variance of the endogenous latent variables,
- pre-specified numbers of segments, and
- Mode C path model constellations

have a negative impact on PLS-GAS's hit rates. In contrast, more complex path models improve the method's performance.

Previous studies revealed the relatively good performance of FIMIX-PLS, which is currently the most recognized PLS-SEM segmentation method. In this study, we further substantiate the good quality of FIMIX-PLS results, which serve as a fair benchmark of the PLS-GAS method. Even though FIMIX-PLS shows equal or slightly better outcomes regarding the hit rates in several data constellations, PLS-GAS meets the expected values of the artificially generated data sets with higher precision. This especially holds when segments are not well separated, or when error variances are high. When there is little error, both methods are generally in close correspondence in terms of parameter recovery. This finding also holds for non-normal data constellations where PLS-GAS's and FIMIX-PLS's performances are highly similar. In light of the practical advantages of FIMIX-PLS in terms of run-time, researchers may be inclined to prefer FIMIX-PLS over PLS-GAS. However, fuzzy segments are more common than the exception in empirical research and researchers can never know a priori whether those situations that justify the use of FIMIX-PLS actually occur in practice. Thus, researchers are well advised to err on the side of caution and use PLS-GAS to address the need to uncover and handle unobserved heterogeneity in PLS-SEM. The empirical application of PLS-GAS to the well-known ACSI model and its comparison with FIMIX-PLS results (as documented in prior research; [Ringle et al. \(2010\)](#)) further substantiates this conclusion, as PLS-GAS achieves higher average R^2 values of the endogenous latent variables in each uncovered segment. Lastly, the REBUS-PLS benchmark failed continuously in our study's computational experiments, making its use inadvisable.

Notwithstanding the usefulness of the PLS-GAS approach, we realize that this study has certain limitations. A key requirement for its general use in applications is linked to

its performance in Mode C path mode constellations. Future research should specifically address this issue and more extensively evaluate the PLS-GAS performance regarding PLS path models that include Mode B measurement models for latent variables. In this regard, the PLS-GAS approach is very flexible, as the procedure allows the integration of other objective functions that, for example, make use of communality and redundancy measures that may better account for Mode B measurement model situations. Consequently, evaluating the effect of different objective criteria on the procedure's performance is another promising field for future research work. Such studies should also aim at improving the GA parameter settings. The systematic exploration of the effects on different parameter settings' segmentation outcome would enhance the applicability of PLS-GAS in research practice.

With the exception of model complexity's positive effect on PLS-GAS's performance, the influence of the design factors on the segmentation results is remarkably consistent with our expectations. Future research should explore this divergence in greater detail by systematically varying the number of constructs and structural model relationships in the model to disentangle the effect of model complexity on PLS-GAS. Another promising area for future research is the specific analysis of PLS-GAS's behavior when the number of segments increases. The data assignment problem becomes much more complex and, in consequence, PLS-GAS's hit rates decrease. Subsequent studies may provide an in-depth analysis of PLS-GAS's data segmentation behavior and may advance towards improving the method's performance in this respect. In this regard a comparison of two novel PLS-GAS and PLS-POS methods, especially regarding the (run-time) performance of the two methods, might also be an interesting research opportunity. Moreover, even though [Becker et al. \(2013\)](#) recently presented a systematic approach for selecting an appropriate number of segments in applications, future research must present solutions to cope with this problem. [Ringle et al. \(2010\)](#) have suggested running FIMIX-PLS on the data first to obtain likelihood-based criteria such as AIC_3 or CAIC to guide this decision. Future research needs to effectively combine both approaches to decide on a suitable number of segments (i.e., by using FIMIX-PLS) and to obtain appropriate segmentation results (i.e., by using PLS-GAS). Finally, future research should more specifically address the problem of matching the PLS-SEM segmentation results by means of explanatory variables to better describe and interpret the solutions and, thus, draw superior conclusions.

Acknowledgments The authors thank the area editor and four anonymous reviewers for their helpful comments as well as the time and care devoted to our research.

References

- Albers S (2010) PLS and success factor studies in marketing. In: Esposito Vinzi V, Chin WW, Henseler J, Wang H (eds) *Handbook of partial least squares: concepts, methods and applications*. Springer Handbooks of Computational Statistics Series, vol II. Springer, Heidelberg/ Dordrecht/London/New York, pp 409–425
- Albers S, Hildebrandt L (2006) Methodische Probleme bei der Erfolgsfaktorenforschung—Messfehler, formative versus reflektive Indikatoren und die Wahl des Strukturgleichungs-Modells. *Z für betriebswirtschaftliche Forsch* 58:2–33
- Anderberg MR (1973) *Cluster analysis for applications*. Academic Press, New York

- Anderson EW, Fornell CG (2000) Foundations of the American customer satisfaction index. *Total Quality Manag* 11:869–882
- Aptech (2008) GAUSS for Windows Version 9.0. Aptech Systems, Black Diamond
- Becker J-M, Rai A, Ringle CM et al (2013) Discovering unobserved heterogeneity in structural equation models to avert validity threats. *MIS Q* (forthcoming)
- Becker J-M, Ringle CM, Völckner F (2009) Prediction-oriented segmentation: a new methodology to uncover unobserved heterogeneity in PLS path models. In: *Proceedings of the 38th Conference of the European Marketing Academy (EMAC)*, Nantes
- Boomsma A, Hoogland JJ (2001) The robustness of LISREL modeling revisited. In: Cudeck R, du Toit S, Sörbom D (eds) *Structural equation modeling: present and future*. Scientific Software International, Chicago, pp 139–168
- Cenfetelli RT, Bassellier G (2009) Interpretation of formative measurement in information systems research. *MIS Q* 33:689–708
- Chin WW (2010) How to write up and report PLS analyses. In: Esposito Vinzi V, Chin WW, Henseler J, Wang H (eds) *Handbook of partial least squares: concepts, methods and applications*. Springer Handbooks of Computational Statistics Series, vol II. Springer, Heidelberg/Dordrecht/ London/New York, pp 655–690
- Chin WW (1998) The partial least squares approach to structural equation modeling. In: Marcoulides GA (ed) *Mod Methods Bus Res*. Erlbaum, Mahwah, pp 295–358
- Chin WW (2003) A permutation procedure for multi-group comparison of PLS models. In: Vilares M, Tenenhaus M, Coelho PS, Esposito Vinzi V, Morineau A (eds) *Focus on customers: Proceedings of the 3rd International Symposium on PLS and Related Methods (PLS'03)*. Decisia, Paris, pp 33–43
- Chin WW, Marcolin BL, Newsted PN (2003) A partial least squares approach for measuring interaction effects: results from a Monte Carlo simulation study and an electronic mail emotion/adoption study. *Inf Syst Res* 14:189–217
- Coltman T, Devinney TM, Midgley DF et al (2008) Formative versus reflective measurement models: two applications of formative measurement. *J Bus Res* 61:1250–1262
- Cowgill MC, Harvey RJ, Watson LT (1999) A genetic algorithm approach to cluster analysis. *Comput Math Appl* 39:99–108
- Diamantopoulos A (2011) Incorporating formative measures into covariance-based structural equation models. *MIS Q* 35:335–A335
- Diamantopoulos A, Riefler P, Roth KP (2008) Advancing formative measurement models. *J Bus Res* 61:1203–1218
- Dijkstra TK (1983) Some comments on maximum likelihood and partial least squares methods. *J Econom* 22:67–90
- Esposito Vinzi V, Ringle CM, Squillacciotti S et al (2007) Capturing and treating unobserved heterogeneity by response based segmentation in PLS path modeling: a comparison of alternative methods by computational experiments. In: ESSEC Research Center, Working Paper No. 07019, Cergy Pontoise Cedex
- Esposito Vinzi V, Trinchera L, Squillacciotti S et al (2008) REBUS-PLS: a response-based procedure for detecting unit segments in PLS path modelling. *Appl Stoch Models Bus Ind* 24:439–458
- Falk RF, Miller NB (1992) *A primer for soft modeling*. University of Akron Press, Akron
- Fogel DB (2006) *Evolutionary computation: toward a new philosophy of machine intelligence*. IEEE Press, Piscataway
- Fornell CG, Bookstein FL (1982) Two structural equation models: LISREL and PLS applied to consumer exit-voice theory. *J Mark Res* 19:440–452
- Fornell CG, Johnson MD, Anderson EW et al (1996) The American customer satisfaction index: nature, purpose, and findings. *J Mark* 60:7–18
- Götz O, Liehr-Gobbers K, Krafft M (2010) Evaluation of structural equation models using the partial least squares (PLS) approach. In: Esposito Vinzi V, Chin WW, Henseler J, Wang H (eds) *Handbook of partial least squares: concepts, methods and applications*. Springer Handbooks of Computational Statistics Series, vol II. Springer, Heidelberg/Dordrecht/London/New York, pp 691–711
- Grün B, Leisch F (2008) Flexmix 2: finite mixtures with concomitant variables and varying constant parameters. *J Stat Softw* 28:1–35
- Gudergan SP, Ringle CM, Wende S et al (2008) Confirmatory tetrad analysis in PLS path modeling. *J Bus Res* 61:1238–1249
- Haenlein M, Kaplan AM (2004) A beginner's guide to partial least squares analysis. *Understand Stat* 3:283–297

- Hahn C, Johnson MD, Herrmann A et al (2002) Capturing customer heterogeneity using a finite mixture PLS approach. *Schmalenbach Bus Rev* 54:243–269
- Hair JF, Ringle CM, Sarstedt M (2011) PLS-SEM: indeed a silver bullet. *J Mark Theory Pract* 19:139–151
- Hair JF, Ringle CM, Sarstedt M (2012a) Partial least squares: the better approach to structural equation modeling? *Long Range Plan* 45:312–319
- Hair JF, Sarstedt M, Pieper TM et al (2012b) Applications of partial least squares path modeling in management journals: a review of past practices and recommendations for future applications. *Long Range Plan* 45:320–340
- Hair JF, Sarstedt M, Ringle CM et al (2012c) An assessment of the use of partial least squares structural equation modeling in marketing research. *J Acad Mark Sci* 40:414–433
- Hair JF, Hult GTM, Ringle CM et al (2013a) A primer on partial least squares structural equation modeling (PLS-SEM). Sage, Thousand Oaks
- Hair JF, Ringle CM, Sarstedt M (2013b) Partial least squares structural equation modeling: rigorous applications, better results and higher acceptance. *Long Range Planning* 46:1–12
- Henseler J, Chin WW (2010) A comparison of approaches for the analysis of interaction effects between latent variables using partial least squares path modeling. *Struct Equ Model* 17:82–109
- Henseler J, Sarstedt (2013) Goodness of fit indices for partial least squares path modeling. *Comput Stat* (forth coming)
- Henseler J, Ringle CM, Sarstedt M (2012) Using partial least squares path modeling in international advertising research: basic concepts and recent issues. In: Okazaki S (ed) *Handbook research in international advertising*. Edward Elgar Publishing, Cheltenham, pp 252–276
- Henseler J, Ringle CM, Sinkovics RR (2009) The use of partial least squares path modeling in international marketing. *Adv Int Mark* 20:277–320
- Holland JH (1975) *Adaptation in natural and artificial systems*. University of Michigan Press, Ann Arbor
- Hui BS, Wold H (1982) Consistency and consistency at large of partial least squares estimates. In: Jöreskog KG, Wold H (eds) *Systems under indirect observation, Part II*. North Holland, Amsterdam, pp 119–130
- Jedidi K, Jagpal HS, Desarbo WS (1997) Finite-mixture structural equation models for response-based segmentation and unobserved heterogeneity. *Mark Sci* 16:39–59
- Jöreskog KG (1982) The LISREL approach to causal model-building in the social sciences. In: Wold H, Wold KG (eds) *Systems under indirect observation, Part I*. North-Holland, Amsterdam, pp 81–100
- Jöreskog KG (1978) Structural analysis of covariance and correlation matrices. *Psychometrika* 43:443–477
- Jöreskog KG, Wold H (1982) The ML and PLS techniques for modeling with latent variables: historical and comparative aspects. In: Wold H, Jöreskog KG (eds) *Systems under indirect observation, Part I*. North-Holland, Amsterdam, pp 263–270
- Lee L, Petter S, Fayard D et al (2011) On the use of partial least squares path modeling in accounting research. *Int J Acc Inf Syst* 12:305–328
- Leisch F, Monecke A (2012) semPLS: structural equation modeling using partial least squares. *J Stat Softw* 48
- Lohmöller J-B (1989) *Latent variable path modeling with partial least squares*. Physica, Heidelberg
- Maulik U, Bandyopadhyay S (2000) Genetic algorithm-based clustering technique. *Pattern Recognit* 33:1455–1465
- Micceri T (1989) The unicorn, the normal curve, and other improbable creatures. *Psychol Bull* 105:156–166
- Michalewicz Z et al (1996) *Genetic algorithms + data structures = evolutionary programs*. Springer, Berlin
- Money KG, Hillenbrand C, Henseler J et al (2012) Exploring unanticipated consequences of strategy amongst stakeholder segments: the case of a European revenue service. *Long Range Plan* 45:395–423
- Mooi EA, Sarstedt M (2011) *A concise guide to market research: the process, data, and methods using IBM SPSS statistics*. Springer, Berlin
- Murthy CA, Chowdhury N (1996) In search of optimal clusters using genetic algorithms. *Pattern Recognit* 17:825–832
- Navarro A, Acedo FJ, Losada F et al (2011) Integrated model of export activity: analysis of heterogeneity in managers' orientations and perceptions on strategic marketing management in foreign markets. *J Mark Theory Pract* 19:187–204
- Peng DX, Lai F (2012) Using partial least squares in operations management research: a practical guideline and summary of past research. *J Oper Manag* 30:467–480
- R Core Team (2012) *R: a language and environment for statistical computing*. In: R Foundation for Statistical Computing, Vienna

- Reinartz WJ, Haenlein M, Henseler J (2009) An empirical comparison of the efficacy of covariance-based and variance-based SEM. *Int J Mark Res* 26:332–344
- Rigdon EE, Ringle CM, Sarstedt M (2010) Structural modeling of heterogeneous data with partial least squares. In: Malhotra NK (ed) *Rev Mark Res*. Sharpe, Armonk, pp 255–296
- Rigdon EE, Ringle CM, Sarstedt M et al (2011) Assessing heterogeneity in customer satisfaction studies: across industry similarities and within industry differences. *Adv Int Mark* 22:169–194
- Ringle CM, Sarstedt M, Mooi EA (2010) Response-based segmentation using finite mixture partial least squares: theoretical foundations and an application to American customer satisfaction index data. *Ann Inf Syst* 8:19–49
- Ringle CM, Sarstedt M, Schlittgen R (2010) Finite mixture and genetic algorithm segmentation in partial least squares path modeling: identification of multiple segments in a complex path model. In: Fink A, Lausen B, Seidel W, Ultsch A (eds) *Advances in data analysis, data handling and business intelligence*. Springer, Berlin-Heidelberg, pp 167–176
- Ringle CM, Sarstedt M, Schlittgen R et al (2013) PLS path modeling and evolutionary segmentation. *J Bus Res* (forthcoming)
- Ringle CM, Sarstedt M, Straub DW (2012) A critical look at the use of PLS-SEM in MIS quarterly. *MIS Q* 36:iii–xiv
- Ringle CM, Schlittgen R (2007) A genetic segmentation approach for uncovering and separating groups of data in PLS path modeling. In: Martens H, Næs T, Martens M (eds) *Causalities explored by indirect observation: proceedings of the 5th international symposium on PLS and related methods (PLS'07)*. MATFORSK, Åas, pp 75–78
- Ringle CM, Wende S, Will A (2010) Finite mixture partial least squares analysis: methodology and numerical examples. In: Esposito Vinzi V, Chin WW, Henseler J, Wang H (eds) *Handbook of partial least squares: concepts, methods and applications*. Springer Handbooks of Computational Statistics Series, vol II. Springer, Heidelberg, pp 195–218
- Ringle CM, Wende S, Will A (2005) SmartPLS 2.0. <http://www.smartpls.de>, Hamburg
- Robins JA (2012) Partial-least squares. *Long Range Plan* 45:309–311
- Roldán JL, Sánchez-Franco MJ (2012) Variance-based structural equation modeling: guidelines for using partial least squares in information systems research. In: *Research Methodologies, Innovations and Philosophies in Software Systems Engineering and Information Systems*. IGI Global, pp 193–221
- Sánchez G, Trinchera L (2013) R Package plspm: tools for partial least squares path modeling (version 0.3.5). <http://cran.r-project.org/web/packages/plspm/>
- Sarstedt M (2008a) A review of recent approaches for capturing heterogeneity in partial least squares path modelling. *J Model Manag* 3:140–161
- Sarstedt M (2008b) Market segmentation with mixture regression models: understanding measures that guide model selection. *J Target Meas Anal Mark* 16:228–246
- Sarstedt M, Becker J-M, Ringle CM et al (2011a) Uncovering and treating unobserved heterogeneity with FIMIX-PLS: which model selection criterion provides an appropriate number of segments? *Schmalenbach Bus Rev* 63:34–62
- Sarstedt M, Henseler J, Ringle CM (2011b) Multi-group analysis in partial least squares (PLS) path modeling: alternative methods and empirical results. *Adv Int Mark* 22:195–218
- Sarstedt M, Ringle CM (2010) Treating unobserved heterogeneity in PLS path modelling: a comparison of FIMIX-PLS with different data analysis strategies. *J Appl Stat* 37:1299–1318
- Sarstedt M, Schwaiger M, Ringle CM (2009) Do we fully understand the critical success factors of customer satisfaction with industrial goods?—extending Festge and Schwaiger's model to account for unobserved heterogeneity. *J Bus Mark Manag* 3:185–206
- Shah R, Goldstein SM (2006) Use of structural equation modeling in operations management research: looking back and forward. *J Oper Manag* 24:148–169
- Sosik JJ, Kahai SS, Piovoso MJ (2009) Silver Bullet or Voodoo statistics? A primer for using the partial least squares data analytic technique in group and organization research. *Group Organ Manag* 34:5–36
- Squillacciotti S (2010) Prediction-oriented classification in PLS path modeling. In: Esposito Vinzi V, Chin WW, Henseler J, Wang H (eds) *Handbook of partial least squares: concepts, methods and applications*. Springer Handbooks of Computational Statistics Series, vol II. Springer, Heidelberg, pp 219–233
- Squillacciotti S (2005) Prediction oriented classification in PLS path modeling. In: Aluja T, Casanovas J, Esposito Vinzi V, Tenenhaus M (eds) *PLS & marketing: Proceedings of the 4th International Symposium on PLS and Related Methods (PLS'05)*. DECISIA, Paris, pp 499–506
- Tenenhaus M, Esposito Vinzi V et al (2005) PLS path modeling. *Comput Stat Data Anal* 48:159–205

- Völckner F, Sattler H, Hennig-Thurau T et al (2010) The role of parent brand quality for service brand extension success. *J Serv Res* 13:359–361
- Wedel M, Kamakura W, Arora N et al (1999) Discrete and continuous representations of unobserved heterogeneity in choice modeling. *Mark Lett* 10:219–232
- Wedel M, Kamakura WA (2000) *Market segmentation: conceptual and methodological foundations*. Kluwer, Boston
- Wold H (1982) Soft modeling: the basic design and some extensions. In: Jöreskog KG, Wold H (eds) *Systems under indirect observations: Part II*. North-Holland, Amsterdam, pp 1–54